Notebook 1: Data Selection,Cleaning and Wrangling

Introduction:

In this notebook, the objective is to prepare the dataset for further analysis by performing data selection, cleaning, and preprocessing. Real-world datasets often contain missing values, inconsistencies, and noise, which must be handled carefully to ensure accurate analysis and modeling.

```
import pandas as pd
import numpy as np
```

1. Data Selection

The dataset selected for this analysis is a student performance dataset. This dataset contains both categorical and numerical variables, making it suitable for data analysis and machine learning tasks. It helps in understanding how different factors influence student performance.

```
import pandas as pd

try:
    # Load dataset with safe encoding
    df = pd.read_csv("student_performance.csv", encoding='latin-1')

    # Display first 5 rows
    print(df.head())

except FileNotFoundError:
    print("Error: File not found. Make sure 'student_performance.csv' is uploaded.")

except Exception as e:
    print("Error:", e)
```

```
Error: File not found. Make sure 'student_performance.csv' is uploaded.
```

2. Understanding the Dataset

Before cleaning, it is important to understand the structure of the dataset, including the number of rows, columns, and data types of each attribute

```
import pandas as pd

try:
    df = pd.read_csv("student_performance.csv", encoding='latin-1')

    # Shape of dataset
    print("Shape of dataset:", df.shape)

    # Column names
    print("\nColumns in dataset:")
    print(df.columns)

    # Dataset information
    print("\nDataset Info:")
    df.info()

except FileNotFoundError:
    print("Error: File not found. Please upload 'student_performance.csv'.")

except Exception as e:
    print("Error:", e)
```

```
Error: File not found. Please upload 'student_performance.csv'.
```

3.Attribute Description

The dataset consists of attributes such as gender (categorical), age (integer), study hours (float), attendance (float), and marks (float). The marks column is considered the target variable, while the others are independent variables.

Attribute	Description	Type	Role
gender	Male/Female	Categorical	Independent
age	Student age	Integer	Independent
study_hours	Weekly study time	Float	Independent
attendance	Attendance %	Float	Independent
marks	Final score	Float	Dependent

4.Handling Missing Values

Missing values are identified and handled to avoid incorrect analysis. Numerical columns are filled using statistical methods such as mean or median.

```
import pandas as pd

try:
    # Load dataset
    df = pd.read_csv("student_performance.csv", encoding='latin-1')

    # Check missing values
    print("Missing values in each column:\n")
    print(df.isnull().sum())

except FileNotFoundError:
    print("Error: File not found. Please upload 'student_performance.csv'.")

except Exception as e:
    print("Error:", e)
```

```
Error: File not found. Please upload 'student_performance.csv'.
```

```
import pandas as pd

try:
    # Load dataset
    df = pd.read_csv("student_performance.csv", encoding='latin-1')

    # Check if columns exist
    if 'study_hours' in df.columns and 'attendance' in df.columns:

        # Convert to numeric (in case of wrong data type)
        df['study_hours'] = pd.to_numeric(df['study_hours'], errors='coerce')
        df['attendance'] = pd.to_numeric(df['attendance'], errors='coerce')

        # Fill missing values
        df['study_hours'].fillna(df['study_hours'].mean(), inplace=True)
        df['attendance'].fillna(df['attendance'].median(), inplace=True)

        print("Missing values handled successfully!")

    else:
        print("Error: Required columns not found in dataset.")

except FileNotFoundError:
    print("Error: File not found. Please upload 'student_performance.csv'.")

except Exception as e:
    print("Error:", e)
```

```
Error: File not found. Please upload 'student_performance.csv'.
```

5.Removing Duplicate Records

Duplicate rows can bias the analysis, so they are identified and removed.

```
import pandas as pd

try:
    # Load dataset
    df = pd.read_csv("student_performance.csv", encoding='latin-1')

    # Check number of duplicate rows
    duplicates = df.duplicated().sum()
    print("Number of duplicate rows:", duplicates)

    # Remove duplicates
    df = df.drop_duplicates()

    print("Duplicates removed successfully!")

except FileNotFoundError:
    print("Error: File not found. Please upload 'student_performance.csv'.")

except Exception as e:
    print("Error:", e)
```

```
Error: File not found. Please upload 'student_performance.csv'.
```

6.Outlier Detection and Removal

Outliers are extreme values that can distort analysis. The Interquartile Range (IQR) method is used to detect and remove outliers.

```
import pandas as pd

try:
    # Load dataset
    df = pd.read_csv("student_performance.csv", encoding='latin-1')

    # Check if column exists
    if 'marks' in df.columns:

        # Convert to numeric (safe handling)
        df['marks'] = pd.to_numeric(df['marks'], errors='coerce')

        # Remove missing values in marks
        df = df.dropna(subset=['marks'])

        # Calculate Q1 and Q3
        Q1 = df['marks'].quantile(0.25)
        Q3 = df['marks'].quantile(0.75)
        IQR = Q3 - Q1

        # Filter out outliers
        df = df[(df['marks'] >= Q1 - 1.5 * IQR) &
                (df['marks'] <= Q3 + 1.5 * IQR)]

        print("Outliers removed successfully!")

    else:
        print("Error: 'marks' column not found in dataset.")

except FileNotFoundError:
    print("Error: File not found. Please upload 'student_performance.csv'.")

except Exception as e:
    print("Error:", e)
```

```
Error: File not found. Please upload 'student_performance.csv'.
```

7.Data Filtering

The dataset is filtered to remove students with very low attendance, as such records may not reflect meaningful academic behavior.

```
import pandas as pd

try:
    # Load dataset
    df = pd.read_csv("student_performance.csv", encoding='latin-1')

    # Check if column exists
    if 'attendance' in df.columns:

        # Convert to numeric safely
        df['attendance'] = pd.to_numeric(df['attendance'], errors='coerce')
```

```
        # Remove missing values
        df = df.dropna(subset=['attendance'])

        # Filter attendance >= 60
        df = df[df['attendance'] >= 60]

        print("Filtering completed: attendance >= 60")

    else:
        print("Error: 'attendance' column not found in dataset.")

except FileNotFoundError:
    print("Error: File not found. Please upload 'student_performance.csv'.")

except Exception as e:
    print("Error:", e)
```

```
Error: File not found. Please upload 'student_performance.csv'.
```

8.Data Transformation

Categorical variables are converted into numerical form using encoding techniques.

```
import pandas as pd

try:
    # Load dataset
    df = pd.read_csv("student_performance.csv", encoding='latin-1')

    # Check if column exists
    if 'gender' in df.columns:

        # Standardize text (avoids case mismatch like male/Male)
        df['gender'] = df['gender'].astype(str).str.strip().str.title()

        # Mapping
        df['gender'] = df['gender'].map({'Male': 0, 'Female': 1})

        print("Gender encoding completed successfully.")

    else:
        print("Error: 'gender' column not found.")

except FileNotFoundError:
    print("Error: File not found. Please upload dataset.")

except Exception as e:
    print("Error:", e)
```

```
Error: File not found. Please upload dataset.
```

9.Feature Engineering

A new feature called study efficiency is created to measure how effectively students use their study time.

```
import pandas as pd

try:
    # Load dataset
    df = pd.read_csv("student_performance.csv", encoding='latin-1')

    # Display statistical summary
    print(df.describe())

except FileNotFoundError:
    print("Error: File not found. Please upload 'student_performance.csv'.")

except Exception as e:
    print("Error:", e)
```

```
Error: File not found. Please upload 'student_performance.csv'.
```

10.Final Dataset Summary

After preprocessing, the dataset is summarized to ensure it is clean and ready for analysis.

```
import pandas as pd

try:
```

```
    df = pd.read_csv("student_performance.csv", encoding='latin-1')

    print(df.describe())

except FileNotFoundError:
    print("Error: File not found. Please upload 'student_performance.csv'.")

except Exception as e:
    print("Error:", e)
```

```
Error: File not found. Please upload 'student_performance.csv'.
```

Conclusion

At the end of this notebook, the dataset has been successfully cleaned and transformed. It is now ready for exploratory data analysis.